

$(\Omega, \mathcal{B}, P)$

Def^m: Let $X_1, X_2, \dots, X_n$ be a sequence of jointly distributed R.V.s (for $n \geq 1$) defined on the same probability space Ω . Let X be another R.V. defined on Ω . we say that X_n converges to X in probability, i.e., $X_n \xrightarrow{P} X$, if for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \underline{P(|X_n - X| > \epsilon)} = 0.$$

$$\epsilon = \frac{1}{10}, \frac{1}{102}, \frac{1}{103}$$

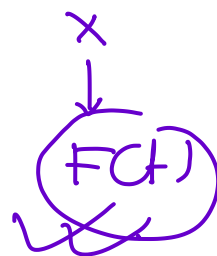
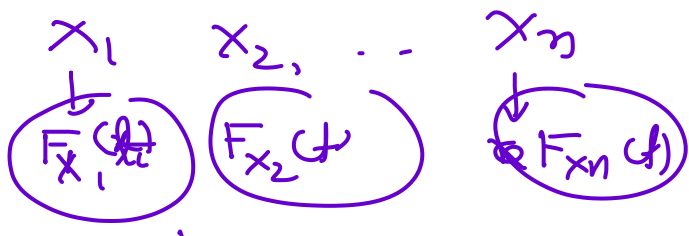
Defⁿ: Convergence in Distribution:

X_n converges to X in distribution or in law

if $\lim_{n \rightarrow \infty} F_{X_n}(t) = F_X(t)$ at each point t , where

$F_X(\cdot)$ is continuous.

$$X_n \xrightarrow{d} X$$



Weak law of Large Numbers (WLLN)

Let $X_1, X_2, \dots, X_n$ are i.i.d (identically independently distributed) random variables such that $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$ are finite.

Then $\bar{X}_n = \left(\frac{1}{n} \sum_{i=1}^n X_i \right) \longrightarrow \mu$ as $n \rightarrow \infty$

• For X_i Bernoulli's, $\left(\frac{1}{n} \sum_{i=1}^n X_i \right)$ is a good estimate of $\mu = p$

$$\left. \begin{aligned} P(X_i = 0) &= 1 - p, \\ P(X_i = 1) &= p, \end{aligned} \right\}$$

$$1 \cdot p + 0 \cdot (1 - p) = \boxed{p}$$